TEACHING NOTES SERIES

How Young Children Learn

Child psychology for the primary classroom — how young minds think, remember, play and grow, and what it means for the way we teach.



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These are practical study notes for primary school teachers, written in simple English. They bring together the main ideas about child psychology and learning that appear in teacher induction books — how children develop, how they think at different ages, and how a teacher can work *with* the child's mind instead of against it.

Every chapter ends with what the idea means in a real classroom, because psychology is only useful when it changes what happens on Monday morning.

Children Are Not Small Adults

The most important idea in child psychology is also the simplest: a young child is not a small, unfinished adult. A child of six does not just know *less* than an adult — she thinks in a completely different way. Her logic, her sense of time, her memory and her feelings all work differently.

When a lesson fails, the reason is often not that the child is lazy or the teacher is weak. The reason is that the lesson was built for an adult mind and given to a child's mind. Good primary teaching starts by asking: how does a child of this age actually learn?

> **Plain definition:** Child psychology is the study of how children grow in mind, feelings, language and behaviour from birth through the school years — and how adults can support that growth.

Five truths about the young learner

- **Children learn by doing.** Touching, moving, building and trying teach more than listening at this age.
- **Children learn through their senses first.** What they can see, hear and handle becomes real; what is only talked about stays foggy.
- **Children learn in small steps.** A new idea must connect to something they already know, or it will not stay.
- **Children learn with feelings attached.** A child who feels safe and liked learns faster. A child who is afraid stops trying.
- **Children develop at different speeds.** Two children of the same age can be a full year apart in readiness, and both are normal.

How a young brain learns

Think of the brain as a city building its roads. Every experience — a word heard, a block stacked, a question answered — lays down or strengthens a connection. Connections that are used again and again become strong roads. Connections that are never used fade away.

This is why **repetition with variety** matters so much in primary classes. The child meets the same idea in a song, a game, a picture and a story, and each meeting makes the road stronger. It is also why the early years matter so much: the primary school years are the years when the most roads are built.

> **In the classroom:** Never ask only, "Have I taught this?" Ask, "Has the child met this idea in more than one way, and used it with his own hands?"

Piaget: Thinking Grows in Stages

The Swiss psychologist Jean Piaget watched children closely for many years and found that thinking develops in stages. Children pass through the stages in the same order, though not at exactly the same age. A teacher who knows the stages stops demanding thinking that a child's stage cannot yet give.

Stage	Age	How the child thinks
Sensorimotor	0-2 years	Learns through the senses and movement — looking, touching, grasping, putting things in the mouth. By the end, the child knows objects still exist when hidden.
Pre-operational	2-7 years	Language and imagination grow fast. Thinking is centred on the child's own view, and appearance wins over logic: the tall thin glass "has more" water. Most children entering primary school are here.
Concrete operational	7-11 years	The child can now think logically — but only about real, touchable things. Sorting, ordering, and understanding that quantity stays the same become possible. This is the main primary school stage.
Formal operational	11+ years	Abstract thinking begins: ideas about ideas, hypotheses, "what if" questions. This belongs mostly to secondary school, not primary.

What this means for a primary teacher

- **Do not rush stages.** A six-year-old who cannot grasp an abstract rule is not dull; her stage has not arrived yet. Pushing harder does not speed the stage up.
- **Teach with real things.** In the concrete stage, logic works when the child can touch the problem. Counters, sticks, stones and bottles teach number better than symbols alone.
- **Expect egocentric answers.** A young child genuinely cannot yet see a problem from another person's side. Teach turn-taking and listening as skills, patiently.
- **Let children discover.** Piaget showed that children build understanding by acting on the world and noticing what happens. A classroom where children only copy gives them little to build with.

> **Key word — schema:** A schema is a mental pattern a child builds from experience, such as "birds fly." New experiences are fitted into old schemas, or the schema itself changes. Teaching is, in large part, the careful changing of schemas.

Vygotsky: Children Learn With Others First

Lev Vygotsky, a Russian psychologist, added the other half of the picture. Piaget showed the child exploring alone; Vygotsky showed that learning is social first. A child learns a skill with help — from a teacher, a parent or a stronger classmate — and only later can do it alone. What a child does with support today, she does by herself tomorrow.



Scaffolding: the teacher supports each step, then steps back as the child climbs alone.

> **Zone of Proximal Development (ZPD):** The gap between what a child can do alone and what the same child can do with good help. Real teaching aims at this zone — not at what is already easy, and not at what is still impossible.

Scaffolding — help that is removed

A scaffold is the temporary frame around a building under construction. In teaching, scaffolding means giving exactly the help a child needs — a hint, a first step done together, a word bank, a worked example — and then removing the help piece by piece as the child grows able. Help that is never removed creates dependence; help that is removed too early creates failure.

- **Model first:** "Watch me do one."
- **Do it together:** "Now we do one together — you tell me the next step."
- **Guide from the side:** "You try; I will only give hints."
- **Step back:** "Now alone. Show me what you can do."

Language sits at the centre of Vygotsky's theory. Talk is not noise in the classroom; talk is thinking out loud. Children who explain, question and argue in pairs are building understanding, which is why well-organised pair and group talk is a learning method, not a reward after learning.

Attention and Memory: Small Tanks, Filled Slowly

Two limits of the young mind shape every primary lesson: attention is short, and working memory is small. A wise teacher plans for both instead of fighting them.

Attention grows with age

A useful rule of thumb used in teacher training: a child can hold focused attention on one task for roughly **two to three minutes per year of age**. A six-year-old manages about 12 to 18 minutes; a ten-year-old, about 20 to 30 minutes. After that, the mind wanders — not from badness, but from biology.

- Change the activity every 10 to 15 minutes in early classes: listen, then do, then talk, then move.
- Put the hardest new learning early in the lesson, while attention is fresh.
- Use short movement breaks. A one-minute stretch often returns five minutes of attention.
- Remove distractions during explanation: hands still, eyes on the speaker, materials closed.

Working memory — the small desk

Working memory is the mind's small desk: the space where a child holds information while using it. An adult holds about four to seven items; a young child holds far fewer, often only two or three. When instructions carry five steps, the desk overflows and the child appears careless or disobedient — when in fact the information simply fell off the desk.

- Give instructions **one or two steps at a time**, and write or draw them where children can see them.
- Ask a child to repeat the instruction back before starting.
- Break big tasks into small visible parts, and tick each part off as it is done.

Long-term memory — built by returning

For learning to stay, it must move from the small desk into long-term memory. Three things make that journey happen: **meaning** (the idea connects to what the child knows), **emotion** (the lesson felt interesting, funny or surprising), and **returning** — meeting the idea again after a day, a week and a month. Psychologists call this spaced revision, and it is far stronger than repeating everything on one day.

> **In the classroom:** Start each lesson with two minutes of "yesterday, last week, last month" questions. This small habit is one of the most powerful memory tools a teacher owns.

Learning Through Play Is Real Learning

In the primary years, play is not a break from learning. Play is one of the main ways learning happens. When children build with blocks, they are meeting shape, balance, counting and problem-solving. When they play shop, they are practising money, language and manners. When they play in a group, they are learning the hardest subject of all: other people.



Blocks, hopscotch and a watering can: number, balance, language and care, all inside play.

What play builds

Kind of play	What it develops
Building and making	Fine hand control, shape and space, planning, patience when things fall
Pretend and role play	Language, imagination, understanding other people's feelings and roles
Games with rules	Turn-taking, fairness, self-control, counting and strategy
Running, climbing, ball games	Body control, health, confidence, reading space and movement
Sand, water and nature play	Early science: pouring, measuring, floating, growing, cause and effect

The teacher's role in play is not to control it, but to **prepare it and enrich it**: choose materials with learning inside them, join for a moment to add a word or a question ("What happens if the tower is wider at the bottom?"), and leave before the play becomes a lesson. Guided play — where the teacher sets up the situation and the child still leads — gives the benefits of play and the direction of teaching together.

Concrete First: Hands, Then Pictures, Then Symbols

One of the most reliable findings in primary teaching is the order in which new ideas should arrive. The psychologist Jerome Bruner described three ways the mind represents the world, and good lessons travel through them in order:

Step	The child meets the idea as...	Example: learning "half"
1. Concrete	Real objects she can touch and move	Cutting a real roti or an apple into two equal parts
2. Pictorial	Pictures and drawings of those objects	Drawing a circle and shading one of two equal parts
3. Abstract	Symbols, words and numbers alone	Writing and using the symbol 1/2

When a child is stuck at the abstract step, the cure is almost never more abstract practice. The cure is to go back one step — to the picture, or to the real object — and rebuild. Skipping the concrete step is the most common cause of the child who can recite a rule but cannot use it.

Language grows the mind

Closely tied to this is language development. Between ages five and eleven, vocabulary grows very fast, and every new word is a new handle for thinking. A child who lacks the word cannot easily hold the idea. This is doubly important in our classrooms, where many children think in their home language and learn in another. Practical bridges help: introduce the new word with a real example, let children explain first in the language they think in, then give them the school-language words for the same idea.

- **Talk before writing.** A child who can say it can usually learn to write it; a child who cannot say it will copy writing without meaning.
- **Word walls and picture labels** turn the classroom itself into a memory aid.
- **Story and rhyme** are language engines: rhythm and repetition make words stay almost without effort.

> **In the classroom:** For every new topic ask, "What will their hands touch first?" If the answer is "nothing," the lesson plan needs one more step at the beginning.

Motivation, Praise and Confidence

Children repeat what feels rewarding and avoid what feels threatening. Motivation psychology gives the primary teacher clear guidance about which feelings to build and which to avoid.

Two kinds of motivation

Intrinsic motivation comes from inside: curiosity, the pleasure of "I did it," interest in the thing itself. **Extrinsic motivation** comes from outside: marks, prizes, stars, fear of punishment. Outside rewards have their place for small routines, but research and classroom experience agree on the order: children who learn for inner reasons keep learning when the prizes stop. The teacher's daily talk is the strongest tool for building the inner kind.

Praise the effort, name the strategy

- **Weak praise:** "You are so clever." — The child learns that ability is fixed, and becomes afraid to risk failing and lose the label.
- **Strong praise:** "You kept trying different ways until the tower stood. That is how builders work." — The child learns that effort and strategy cause success, and tries harder next time.

This is the heart of the **growth mindset** idea associated with Carol Dweck: children who believe ability grows with effort choose harder tasks and recover from mistakes. Children who believe ability is fixed protect their image by avoiding challenge. The difference is often created by nothing more than the words adults use after success and after failure.

Protecting confidence

- **Make success reachable, then stretch it.** A task just above the child's level, inside the ZPD, produces the best feeling in learning: "It was hard, and I did it."
- **Never compare children publicly.** Comparison teaches the weaker child that school is a place where she loses.
- **Treat mistakes as information.** Say, "Good — now we can see exactly what to fix," and mean it.
- **Be careful with fear.** Fear may produce silence; it does not produce learning. A frightened child's brain is busy surviving, not connecting ideas.

> **Plain definition:** Self-efficacy is a child's belief, "I can do this kind of task." It is built mainly from real small successes — which is why the teacher's job is to engineer them, honestly and often.

Every Child Learns Differently

No two children in a class are at the same point in everything. Differences in pace, readiness, language, health, home experience and interest are not problems in the class — they are the normal condition of every class. Fair teaching does not mean giving every child identical treatment. It means giving every child what that child needs to reach the same high goals.

The main differences a teacher will meet

Difference	What the teacher does
Pace of learning	Same goal, different time. Fast finishers get deeper questions, not just more of the same work.
Readiness	Check what a child already knows before teaching the next step; fill gaps early and quietly.
Preferred way in	Some children enter through seeing, some through hearing, some through doing. Offer all three doors to the same idea.
Language background	Allow thinking and first answers in the home language; build the school language on top of understanding, not instead of it.
Confidence and temperament	Shy children may know more than they show. Give low-risk ways to answer: pairs first, writing first, choosing between options.
Additional needs	Some children need glasses, a seat near the front, larger print, extra time, or tasks in smaller pieces. Small adjustments, made early, change a school life.

Multiple intelligences — a wider view of "clever"

Howard Gardner proposed that intelligence is not one single ability but several: language, number and logic, music, body control, understanding nature, understanding people, and understanding oneself, among others. The theory is debated by psychologists, but its classroom lesson is sound and useful: a child who is weak in written work may be strong in building, drawing, leading or remembering through rhythm. A wise teacher watches for each child's strength and uses it as a bridge into the child's weak areas — the strong singer learns tables through rhythm; the strong builder learns measurement through making.

> **In the classroom:** Keep a simple note for each child: "Strong in ___. Needs help in ___." Review it monthly. Teaching from evidence about real children beats teaching from assumptions about an average child who does not exist.

Watching Learning Happen: The Teacher as Observer

Tests show learning after it has happened. Observation shows learning while it is happening, when the teacher can still help. Child psychology becomes daily practice through skilled watching and listening. The signs below are what developing understanding actually looks like in a primary classroom.

Signs that learning is taking root

- The child asks questions that were not in the book — "what if" questions show a schema changing.
- The child uses a new word correctly in her own sentence, not only in repetition.
- The child can explain the idea to a classmate in simpler words.
- Errors change quality: from random guesses to near-misses that show partial understanding.
- The child needs less help than last week on the same kind of task — the scaffold is working.
- The child chooses the activity again in free time. Voluntary return is motivation made visible.

A weekly observation checklist

Tick what you genuinely saw this week, for a few children at a time
☐ Child explained an idea in his or her own words
☐ Child connected today's lesson to something already known
☐ Child persisted after a first failure instead of giving up
☐ Child worked with less support than before (scaffold removed)
☐ Child helped a classmate understand, using simple language
☐ Child showed a strength outside written work (building, drawing, leading, rhythm)
☐ Child who is usually silent found a safe way to take part

Observe a few children deeply each week rather than all children shallowly. Over a term, the notes build a true picture of every learner — far richer than marks alone — and they point directly to the next teaching step for each child.

One-Page Revision Summary

Idea	In one line	In the classroom
Children are not small adults	Young minds think, attend and remember differently	Plan for the child's mind, not the adult's
Brain as building roads	Used connections grow strong; unused ones fade	Repeat key ideas in varied ways, over time
Piaget's stages	Thinking develops in a fixed order; primary years are the concrete stage	Real objects before rules; do not rush stages
Vygotsky's ZPD	Children do with help today what they do alone tomorrow	Teach in the zone; pair and group talk is learning
Scaffolding	Temporary help, removed piece by piece	Model, do together, guide, step back
Attention	About 2-3 minutes per year of age	Change activity every 10-15 minutes; hard work first
Working memory	A very small desk in young children	One or two instruction steps at a time, shown visually
Spaced revision	Returning beats repeating	Begin lessons with yesterday / last week / last month
Play	A main route to learning, not a reward after it	Prepare rich play; guide lightly; join, add a question, leave
Concrete - Pictorial - Abstract	New ideas travel hands, pictures, symbols	When stuck on symbols, go back a step
Motivation	Inner reasons outlast prizes and fear	Praise effort and strategy by name; never compare publicly
Individual differences	Different pace and strengths are normal	Same goals, different support; use strengths as bridges
Observation	Learning is visible while it happens	Watch a few children deeply each week; note the next step

If these notes leave you with one sentence, let it be this: **children learn best when teaching matches how their minds actually work** — actively, socially, concretely, in small steps, with returning practice, inside a classroom where it is safe to try. Everything else in primary teaching is built on that foundation.

These notes are a companion to the other titles in this series, which cover classroom arrangement, teaching methods, classroom management and discipline, and lesson planning for primary classes.